Step-Wise Multiple Linear Regression Model Development for Shrinkage Strain Prediction of Alkali Activated Binder Concrete

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Abstract Among the recent discoveries for alternatives of portland cement (PC) concrete, Alkali-activated binder (AAB) concrete is prolifically being considered as the most eco-friendly and sustainable alternative. The present study evaluates the shrinkage behaviour for three different AAB mixtures containing fly ash and/or slag at different proportions which are activated by sodium hydroxide and sodium silicate. Multiple linear regression models are developed to predict shrinkage strains of ambient-cured AAB concrete as a function of age and percentage of fly ash in the precursor. The aim of this work is to come up with a generalized equation that can predict the shrinkage of various binary blended AAB mixes cured at room temperature. The predicted models are ranked based on RMSE and then compared with the experimental data. The correlations were found to be quite satisfactory ($R^2 = 0.937$) and can be used to estimate the shrinkage for similar AAB mixtures. It is observed that the proposed model agrees more closely with the experimental results from the present study.

Keywords Alkali-activated concrete · AAB · Ambient-cured · Shrinkage strain · Regression model

1 Introduction

Alkali-activated binders (AAB) are one of the most popular *ecolonomical* (Ecological + Economical) alternatives for portland cement (PC). It has been reported by several researchers that AAB has better mechanical properties than PC [1–3]. This is due to polymerization reaction of AAB upon the activation of industrial wastes such

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as fly ash and slag with alkaline solution containing sodium silicate and sodium hydroxide. The polymerization leads to the formation of a stronger aluminosilicate hydrate matrix compared to the calcium silicate hydrate formed by portland cement hydration. The alkali-activation of fly ash takes place at a slow rate under ambient conditions and consequently, thermal curing is preferred. Hence, to make this kind of binders more practical and more energy efficient ambient cured AAB mixes are prepared by partial replacement of fly ash with slag. This is due to the higher reactivity of slag compared to fly ash under ambient conditions. Though the slag-based ambient-cured AAB mixes have shown promising results in improving the mechanical and fire resistance properties compared to PC mixes, the durability properties like shrinkage and creep have not been well established. To overcome this issue, binary and tertiary blended AAB mixes are investigated.

Most of the research on AAB shrinkage is reported either using mortar or paste samples [3–6]. Moreover, the majority of these studies use thermal-cured mixes [7]. Limited information is available on the shrinkage characteristics of AAB concrete cured under ambient conditions. There are various PC shrinkage models namely EN 1992-1-1, B3, ACI 209, Model Code 2010, GL 2000. Most of these models are complex and they take into consideration various parameters like slump of the concrete, relative humidity, compressive strength value and various coefficients which have to be determined based on the type of the mix. Existing models to predict shrinkage for PC are not suitable for AAB concrete [8–11]. 10–20% mass replacement of fly ash with slag resulted in a decrease in shrinkage with increase in slag content for ambient cured AAB mixes over a period of 180 days [1]. Another study carried on binary blended AAB reported that shrinkage increases with increasing slag content and decreases with increasing the Ms modulus (i.e. SiO_2/equivalent Na_2O in the activating solution) [12, 13]. Evidently, there is no conclusive report on the variation of shrinkage with increasing slag content. Hence, the present study focuses on binary blended ambient-cured AAB mix and its shrinkage properties. The objective is to use multiple linear regression to generate a relatively simple model to predict the shrinkage strain of AAB concrete by using age of the specimen and percentage of fly ash in the mix as independent variables (IN) and shrinkage strain as the dependent variable (DV). This would not only make the model less complex but also unique for all binary blended AAB mixes. The materials and methods used in this study are presented in the next section.

2 Materials and Methods

2.1 Materials and Mix Proportions

The materials used for this study are Class F fly ash (FA) and Ground Granulated Blast Furnace Slag (GGBS). These materials comply with the guidelines of ASTM C618 [14] and ASTM C989 [15] respectively. FA is obtained from a thermal power plant in

Ramagundam in the state of Telangana, India. GGBS is obtained from JSW cements, Telangana. Alkali-activating solution is prepared by adding sodium hydroxide pellets to sodium silicate solution. Sodium hydroxide used conforms to food grade with 99% purity and the sodium silicate solution is of industrial grade containing 29.5% SiO_2 and 14.7% Na_2O by weight. Both of these chemicals are obtained from Hychem Chemicals limited. For all the mixes the activating solution was prepared a day before casting. Granite aggregates of nominal size 10 mm that are obtained from local quarries in Hyderabad and 4.75 mm passing local river sand are used for the preparation of concrete. Both comply with the ASTM C33/C33M-18 [16] standard. Chemical admixture used for this work is commercially available polycarboxylate ether based high range water reducer supplied by BASF chemicals according to the guidelines of ASTM C494 [17]. Normal tap water is used for the additional water required in the mix.

2.2 Method

Three different mixes of ambient cured AAB concrete with varying FA to slag ratios, FS0 (100:0), FS30 (70:30) and FS50 (50:50) are used in this study as shown in Table 1. The water to solids ratio (w/s) is maintained at 0.3 for each mix and Ms modulus is taken to be 1.4. Addition of water and superplasticiser vary according to the workability required for a slump of 150 mm. For measuring AAB concrete shrinkage, prisms of dimension 280 × 25 × 25 mm tested in accordance with the specifications of ASTM C157/157M [18] the width of the specimen is taken as 25 mm as the maximum size of aggregate in the concrete is 10 mm. Shrinkage strains (SS) were then calculated as the ratio of change in length to the initial length. The length change of the specimen at any age is calculated using the following equation.

$$\Delta \mathrm{L} = (\mathrm{CRD} - \mathrm{initial}\,\mathrm{CRD})$$
$$SS = \Delta \mathrm{L}/\mathrm{G} \tag{1}$$

Table 1 Details of AAB Concrete mix

Materials	FS 0	FS 30	FS 50
FA/slag	100/0	70/30	50/50
FA (kg/m^3)	400	280	200
Slag (kg/m^3)	0	120	200
Activator solution (kg/m^3)	140	140	140
Coarse aggregate (kg/m^3)	1209	1209	1209
Fine aggregate (kg/m^3)	651	651	651
Water (kg/m^3)	67.5	67.65	67.65
SP (l/m^3)	0	3.14	4

where

ΔL = length change of specimen at any age,

CRD = difference between the comparator reading of the specimen and the reference bar at any age, and

G = the gauge length [280 mm].

The regression analysis and model formation are performed using the R 3.1.3 statistical software package. The procedure followed for arriving at a model equation is explained as follows [19]:

- Data Assimilation: Complete data available is tabulated in one file which can be accessible to the user.
- Data Separation: The complete dataset is randomly divided into training and testing set in a ratio of 70:30 by using "caTools" library in R.
- Model assumption and Validation: Since there are two independent variables and one dependent variable, a linear relationship is assumed initially i.e. $DV = a_0 + a_1*IN_1 + a_2*IN_2$ and the coefficients a_0, a_1 and a_2 are obtained using 'lm' function in R. After finding the relationship between the variables, the statistical significance of each variable is calculated using 't test' which can be obtained by using 'summary' function in R. The insignificant variables are eliminated and the regression analysis is performed again. Subsequently, the statistically significant variables and their relationship with the DV are attained. The most efficient model is selected based on the least root mean square error (RMSE) value. For large RMSE values, then the proposed model form is modified by increasing the polynomial power i.e. $DV = a_0 + a_1*IN_1 + a_2*IN_2 + a_3*IN_1^2 + a_4*IN^2 + a_5*IN_1*IN_2$. Statistically insignificant terms are dropped from the model and the RMSE is computed again to arrive at an optimal model form.

3 Results

A combined set of data containing shrinkage strains, age and percentage of FA readings of all the three mixes is taken as a dataset. The data is divided into training and testing set with 70:30 ratio as explained above and the regression analysis is performed. Five different models are established. These model performances are evaluated and ranked based on the RMSE value obtained from the testing set data. The details of the models with the RMSE value are presented in Table 2. The observed and predicted value of maximum shrinkage strain using the best model is tabulated in Table 3.

Here '*t*' represents the age in days and F represents percentage of FA content in the mix and SS represents shrinkage strain.

Model 5 has the least RMSE value of 58.51 and R^2 of 0.937 and is used for the prediction of shrinkage strains of AAB concrete.

The relationships between the predicted and experimental shrinkage of AAB concrete are presented in Fig. 1. The experimental values reveal that the strains

Table 2 Model equations

Name	Equation	RMSE	R^2
Model 1	$SS = 1.885 * t + 4.84 * F$	215.6532	0.91
Model 2	$SS = 5.59 * 10^2 + 6.64 * t - 1.73 * 10^1 * F - 1.4 * 10^{-2} * t^2 + 1.437 * 10^{-1} * F^2$	157.1211	0.801
Model 3	$SS = 13.15 * t - 6.199 * 10^{-2} * t^2 + 9.043 * 10^{-5} * t^3 - 1.096 * 10^{-1} * F^2 + 1.201 * 10^{-3} * F^3$	123.842	0.9771
Model 4	$SS = 7.64 * t - 0.013 * t^2 + 0.0209 * F^2 - 79.166 * t/F$	101.046	0.961
Model 5	$SS = -35.04 + 23.43 * t - 0.06061 * t^2 + 8.875 * 10^{-5} * t^3 - 1333 * t/F + 38810 * t/F^2$	58.5119	0.937

Table 3 Observed and predicted readings of shrinkage strain observed at 250 days

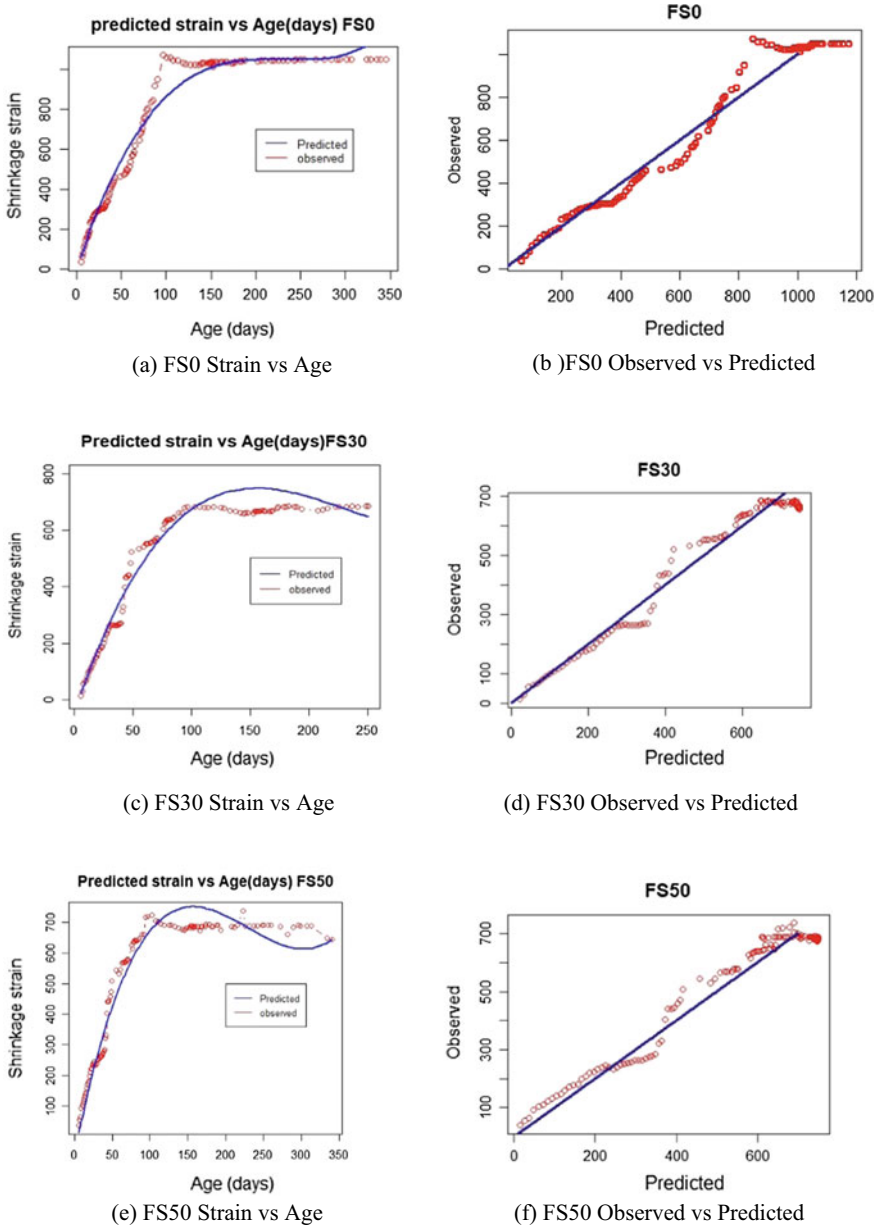
Mix	Observed maximum strain at 250 days (micro strain)	Predicted maximum strain at 250 days (micro strain)
FS 0	1046.205	1036.554
FS 30	685.524	618.4517
FS 50	684.821	615.5538

increase at a relatively swift rate over the first 90 days and then gradually become constant with time up to an age of 365 days. The graphs plotted are for model 5, since the RMSE value is the least for that model. It is evident from the results that the proposed model closely follows the experimental data for all the three AAB mixes and satisfactorily predicts its shrinkage. Hence, unlike previous models containing several IN, the proposed model adequately predicted the shrinkage strains of AAB concrete using a significant mix proportion parameter, i.e. FA proportion in the mix.

4 Conclusions

In the present study, three different AAB concrete mixes are examined for its shrinkage characteristics. Of various multiple linear regression models developed, the model with the least RMSE is used to predict the shrinkage of AAB concrete. The conclusions from the present study are as follows:

- Compared to other models, model 5 has the least RMSE value and is proposed to predict the shrinkage strains of binary blended ambient cured AAB concrete mixes.
- The RMSE and R^2 value of model 5 are 58.51 and 0.937 respectively.
- The shrinkage strains of AAB concrete increase at a rapid rate until 90 days. After 90 days, the increase in shrinkage strain continue at a slower rate.



(a) FS0 Strain vs Age

(b)FS0 Observed vs Predicted

(c) FS30 Strain vs Age

(d) FS30 Observed vs Predicted

(e) FS50 Strain vs Age

(f) FS50 Observed vs Predicted

Fig. 1 Experimental and predicted shrinkage of AAB concrete

- It is observed that shrinkage of AAB concrete decreased with increasing slag content. FS 0 exhibits higher shrinkage, whereas FS 30 and FS 50 exhibit almost similar shrinkage strains.
- The predicted shrinkage of AAB concrete using model 5 follow the same trend as experimental data for mixes with varying fly ash/slag proportion.
- The next phase of this study will compare the shrinkage strains of concrete mixes made from AAB and PC having similar compressive strength, to encourage the practical use of AAB concrete.

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